

CoreOps: Week 3 Module

Kubernetes Security & Platform Hardening

Objective: Master Kubernetes security fundamentals, pod-level & node-level isolation, RBAC correctness, secrets management, and supply-chain trust; the backbone of production-grade clusters.

THURSDAY: Secrets Management

What You Should Understand Today

- What Kubernetes Secrets are
- Why base64 is NOT encryption
- What KMS encryption means
- Why GitOps requires encrypted secrets

Key Outcomes

- You know how to create a secret
 - You know how to mount it into a pod
 - You understand why secrets must be encrypted at rest
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WHAT TO PRACTICE

Lab 1: Create a Secret

Using kubectl create secret.

Lab 2: Mount Secret into a Pod

Use env vars or volume.

Lab 3: Enable Encryption-at-Rest (High Level)

Understand the steps conceptually.

WHERE TO STUDY

- Kubernetes Secrets docs
 - Basic KMS explanation articles
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INFRATHRONE NOTES

- Secrets are the easiest way to leak production access.
 - Keep them encrypted everywhere: cluster, code, Git.
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